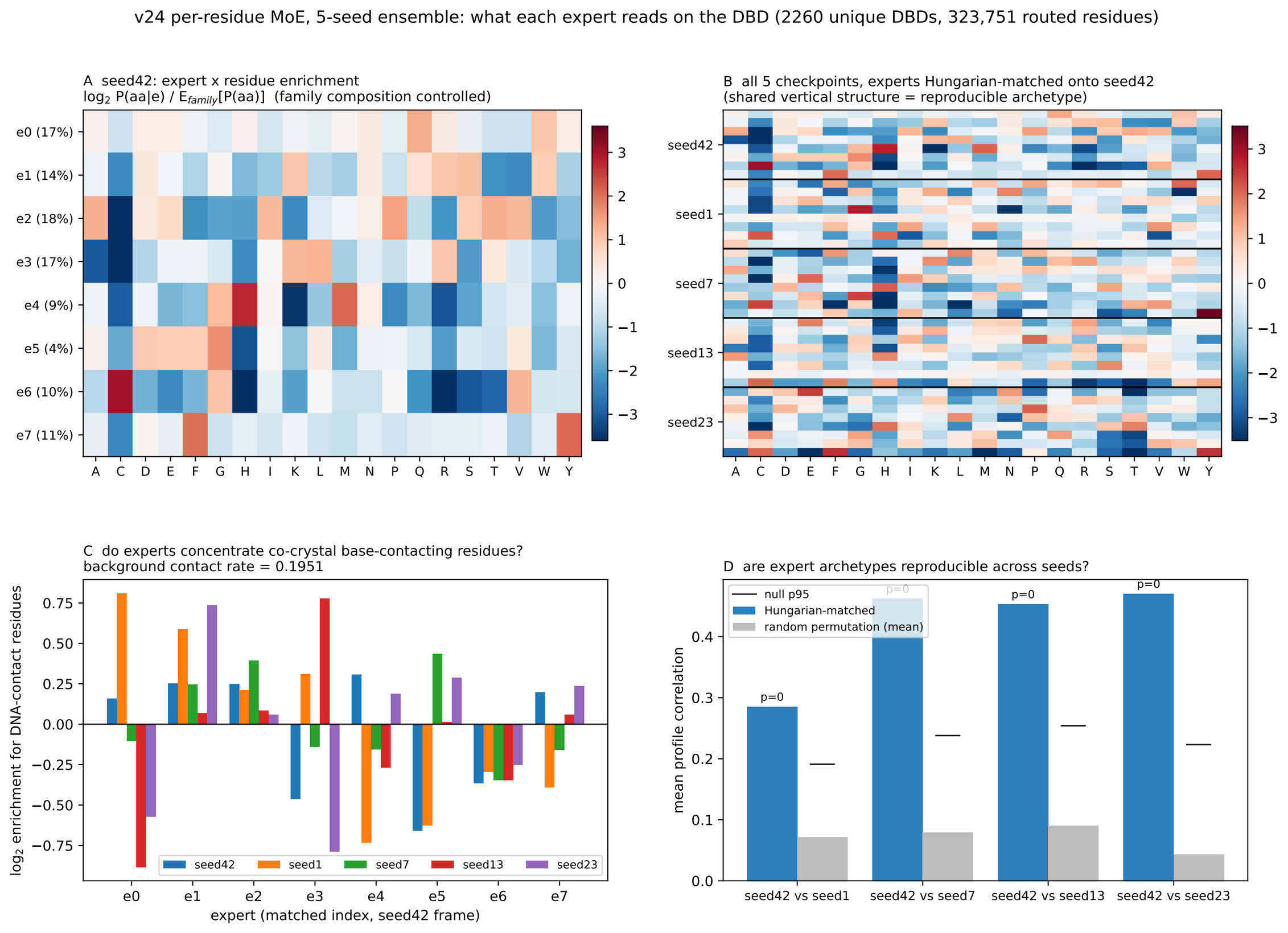
v24 per-residue MoE, 5-seed ensemble: what each expert reads on the DBD (2260 unique DBDs, 323,751 routed residues)
A seed42: expert x residue enrichment
log2 P(aa|e) / E_family[P(aa)] (family composition controlled)
e0 (17%)
e1 (14%)
e2 (18%)
e3 (17%)
e4 (9%)
e5 (4%)
e6 (10%)
e7 (11%)
A C D E F G H I K L M N P Q R S T V W Y
3 2 1 0 −1 −2 −3
B all 5 checkpoints, experts Hungarian-matched onto seed42
(shared vertical structure = reproducible archetype)
seed42
seed1
seed7
seed13
seed23
A C D E F G H I K L M N P Q R S T V W Y
3 2 1 0 −1 −2 −3
C do experts concentrate co-crystal base-contacting residues?
background contact rate = 0.1951
log2 enrichment for DNA-contact residues
0.75 0.50 0.25 0.00 −0.25 −0.50 −0.75
seed42 seed1 seed7 seed13 seed23
e0 e1 e2 e3 e4 e5 e6 e7
expert (matched index, seed42 frame)
D are expert archetypes reproducible across seeds?
null p95
Hungarian-matched
random permutation (mean)
mean profile correlation
0.0 0.1 0.2 0.3 0.4
p=0
p=0
p=0
p=0
seed42 vs seed1
seed42 vs seed7
seed42 vs seed13
seed42 vs seed23